A-A-51185
January 9, 1991

COMMERCIAL ITEM DESCRIPTION

BLADES, BAND SAW, COMPOSITE STEEL, METAL CUTTING, MATRIX HIGH SPEED STEEL

The General Services Administration has authorized the use of this commercial item description.

Abstract. This description covers composite steel band saw blades with the cutting edges manufactured of matrix high speed steel. The blades shall be of the standard, commercial type that are used for sawing ferrous and non-ferrous metals in a production environment.

Classification. The blades shall be of the following styles. The style (tooth form), toothset, blade width, blade thickness, pitch, coil stock or welded blade, length, length tolerance, and quantity to be furnished shall be as specified (see ordering data).

Style A - Standard tooth form.

Style B - Hook tooth form.

Style C - Variable tooth form.

Salient characteristics.

1. Design. The blades shall be new and of the manufacturer's commercial design in effect on the date of the solicitation. The band saw blade shall have one toothed cutting edge of matrix high speed steel welded to a backing of low alloy spring steel. The gullets of the teeth shall extend below the weld.

2. Material. The welded cutting edge of the blades of which the teeth are formed shall be made from matrix high speed steel. The blade backing shall be produced from a low alloy spring steel. All materials shall conform to ASME B94.51M.

Beneficial comments recommendations, additions, deletions, clarifications, etc., and any data which may improve this document should be sent to: DefenseIndustrial Plant Equipment Center, ATTN: DIPEC-SSG, 2163 Airways Blvd., Memphis, TN 38114-5051.

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3. Tooth form. With the exception of the variable tooth form, tooth forms shall conform to ASME B94.51M. The variable tooth form, see figure 1, is defined as a standard tooth with a variation of the number of teeth per inch.



FIGURE 1. Variable tooth form.

4. Blade toothset. Unless otherwise specified (see ordering data), the toothset of the blade to be furnished shall be selected from table I. The raker toothset shall conform to ASME B94.51M. The modified raker toothset shall be defined as a transverse setting of individual teeth as illustrated in figure 2, one to the right, one to the left, one to the right, one to the left, and one straight. More or less teeth set to the right or left may be provided in accordance with the toothset configurations currently available in the commercial market.

TABLE I. Blade toothset.

Toothset designation	Blade toothset
R	Raker
M	Modified Raker
W	Wavy

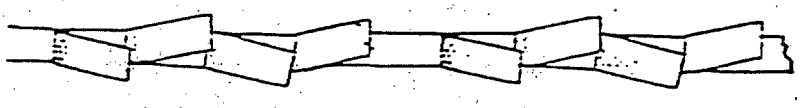


FIGURE 2. Modified raker toothset.

5. Blade width. Unless otherwise specified (see ordering data), the width of the blade to be furnished shall be selected from table II. The tolerance on blades of 3/4 inch widths and less shall be in conformance with ASME B94.51M. The tolerance on 1.000 inch width blades shall be not less than

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0.020 inches, and not more than 0.083 inches. The tolerance on 1.250 inch width blades shall be not less than 0.020 inches, and not more than 0.110 inches. The tolerance on 1.500 inch width blades shall be not less than 0.020 inches, and not more than 0.135 inches. The tolerance on 2.000 inch width blades shall be not more than 0.145 inches. The width of the blade supplied shall be consistent on each blade within ±0.020 inches.

TABLE II. Blade width.

Width designation	Blade width (in.)	Width designation	Blade width (in.)
1	0.250	5	1.000
2	0.375	6	1.250
3	0.500	7	1.500
4	0.750	8	2.000

6. Blade thickness. Unless otherwise specified (see ordering data), the thickness of the blade to be furnished shall be selected from table III. The thickness tolerance of blades 0.024 inch and less shall be ±0.0010 inch. The thickness tolerance of blades 0.050 inch and greater shall be ±0.0015 inch. The thickness tolerance of all other blades shall conform to ASME B94.51M.

TABLE III. Blade thickness.

Thickness designation	Blade thickness (in.)
A	0.020
B	0.025
C	0.035
D	0.042
E	0.050

7. Pitch. Unless otherwise specified (see ordering data), the pitch (teeth per inch) of the blade to be furnished shall be selected from table IV.

8. Blade length. The blades shall be furnished in coils or welded lengths as specified by the procuring activity (see ordering data). The length of the coil or the welded length shall be as specified (see ordering data). The tolerance of the blade length shall be specified (see ordering data). When coils are supplied, there shall be no more than two welds per coil. When welded lengths are supplied, there shall be no more than one weld per blade.

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TABLE IV. Blade pitch.

Style	Pitch designation	Blade pitch	Pitch tolerance (inches)	
			minimum	maximum
A	03	3	.300	.366
	04	4	.225	.275
or	06	6	.150	.184
	08	8	.112	.138
B	10	10	.090	.110
	14	14	.064	.078
	18	18	.050	.062
	24	24	.038	.046
	23	2-3	.300	.600
	34	3-4	.225	.367
	46	4-6	.150	.275
	58	5-8	.112	.220
	59	5-9	.100	.220
C	69	6-9	.100	.183
	60	6-10	.090	.183
	71	7-11	.082	.157
	82	8-12	.075	.138
	83	8-13	.069	.138
	94	10-14	.064	.110

9. Hardness. The blades shall have a body hardness of not less than Rockwell "C" 37 and not greater than Rockwell "C" 52. Blade tooth hardness shall be not less than Rockwell "C" 65 and not greater than Rockwell "C" 68.

10. Marking of blades. The blades shall be tagged, labeled, or marked to identify the blade and manufacturer.

11. Performance. Each band saw blade shall be capable of meeting the saw requirements specified in table V. The test material shall be AISI 1018 cold drawn steel bars, 1x3 inch cross section. The sawing test shall be performed with a blade tension as recommended by the power metal cutting band saw manufacturer for the specific size band saw blade being procured. The sawing test shall be performed on a power metal cutting band saw as appropriate for the specific style and size band saw blade being procured.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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TABLE V. Band saw blade sawing requirements.

Pitch (number of teeth per inch)		Number of sample workpieces	Blade cutting speed (ft. per min.)	Time of cut (seconds)	
Standard or hook	Variable pitch			Break in	Test
-	2-3	6	270	120-138	60-69
3	3-4	3	310	66-76	33-38
4	4-6	2	320	60-69	30-35
6	5-8	1	330	44-51	22-26
-	5-9	1	330	46-54	23-27
-	6-9	1	330	58-67	29-34
8	6-10	1	330	62-71	31-36
-	7-11	1	330	64-74	32-37
10	8-12	1	330	66-76	33-38
-	8-13	1	330	70-80	35-40
-	10-14	1	330	76-88	38-44
14	-	1	330	76-88	38-44
18	-	1	330	86-99	43-50
24	-	1	330	110-140	55-70

NOTE: Sample workpieces shall be positioned side by side resting on the one inch dimensional edge.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this commercial item description are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request shall be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. In accordance with section 23.403 of the Federal Acquisition Regulations, the Government's policy is to acquire items composed of the highest percentage of recovered materials practicable, consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing suppliers' employees to undue hazards from the recovered materials.

Environmental protection. During the manufacture of the blades, materials hazardous to the ecological system as prohibited by the Federal, state, or local statutes in effect on the date of the contract shall not be used or emitted.

Preservation, packaging, packing, labeling and marking. Preservation, packaging, packing, labeling, and marking shall be as specified in the contract order.

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Quality assurance.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

Responsibility for compliance. All items shall meet the requirements specified herein. The inspection set forth in this description shall become a part of the contractors overall inspection system or quality program. The absence of any inspection requirements in the description shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Sampling inspection. The supplier shall randomly select the number of samples in table VI that is applicable to the quantity of items being purchased. Each sample item shall be inspected to determine conformance with the examination and the dimensional inspection requirements of this description. Any sample failing to meet all requirements specified herein shall be considered a defective item. If the number of defective items is equal to or less than the quantity specified under the "Acceptance, number defective" column of table VI, the supplier shall replace the defective samples with samples that meet all the requirements specified herein, the entire lot shall be considered acceptable for shipment. If the number of defective samples is equal to or greater than the quantity specified in the "Rejection, number defective" column of table VI, the entire lot shall be considered defective and unsuitable for acceptance. A lot shall consist of all the blades of one style, width, thickness, pitch, and toothset manufactured under essentially the same conditions and all units processed at the same time and presented for acceptance at one time.

Examination. Each blade submitted for sample inspection shall be visually examined to determine conformance with all requirements of this description.

Dimensional test. Each blade submitted for sample inspection shall be tested for conformance with the dimensional requirements of the salient characteristics.

Hardness test. A hardness test shall be performed in accordance with ASTM E-18 to determine conformance with the hardness requirements of the salient characteristic number 9.

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TABLE VI. Sampling plan.

Quantity of items being purchased	Sample size	Acceptance number defective	Rejection number defective
2 - 8	2	0	1
9 - 15	3	0	1
16 - 25	5	0	1
26 - 50	8	1	2
51 - 90	13	1	2
91 - 150	20	2	3
151 - 280	32	3	4
281 - 500	50	5	6
501 - 1,200	80	8	6
1201 - 3,200	125	12	13
3201 - 10,000	200	18	19

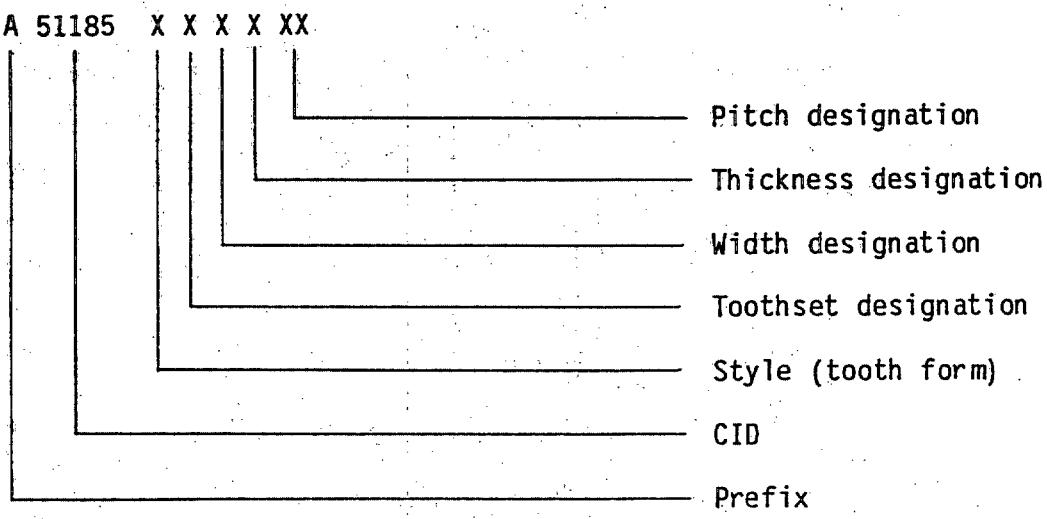
Performance test. Each blade selected for test (see table VI for sample size) shall be tested for conformance with the requirements of table V. The test samples shall be 1x3 inch flat bars of ANSI 1018 cold drawn steel. The sawing test shall be performed on a metal cutting band saw with sufficient horsepower to test the specific blade being procured. A full flow or mist type coolant system shall be used where applicable. The band saw blade cutting speed shall be set at the speed listed in table V under the column heading "Blade cutting speed". Before the test is conducted, five cuts shall be made within the time limits specified in table V under the column with the heading "Break in" time. During the next five cuts, the feed pressure shall be gradually increased so that on the tenth cut the time shall be within the limits specified in table V in the column with the heading "Test". Without further adjustment to the saw, ten cuts shall be made within the time limits specified in table V under the column with the heading "Test".

Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

a. Title, number, and date of this commercial item description
b. Style (tooth form)
c. Toothset
d. Blade width
e. Blade thickness
f. Pitch
g. Coil stock or welded blade
h. Length
i. Length tolerance
j. Quantity

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CID based part identification number (PIN). The following part identification numbering procedure is for Government purposes and does not constitute a requirement for the contractor. Where applicable, Arabic numbers shall be substituted for Roman numerals. Refer to the salient characteristics for origination of the correct letters or numbers to complete the PIN.



Cross reference of classification changes. Classification changes from the previous issue are as follows:

A-A-51134A	A-A-51185
A-A-51134A - Type III - Style B (deleted)	See A-A-51185
A-A-51134A - Type III - Style C (deleted)	See A-A-51185
A-A-51134A - Type III - Style C (deleted)	See A-A-51185

Federal Standard FED-STD-376 is available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5099.

ASME B94.51M is available from the American Society of Manufacturing Engineers, United Engineering Center, 345 East 47th Street, New York, NY 10018-3363.

ASTM E-18 is available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.

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MILITARY INTEREST:

Custodians

Army - AL
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Review Activities

Navy - YD
Air Force - 84
DLA - GS

User Activities

Navy - MC, SH

CIVIL AGENCY COORDINATION ACTIVITY:

GSA - FSS

PREPARING ACTIVITY

DLA - IP

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-51185	2. DOCUMENT DATE (YYMMDD) 91 01 09

3. DOCUMENT TITLE
BLADES, BAND SAW, COMPOSITE STEEL, METAL CUTTING, MATRIX HIGH SPEED STEEL

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Industrial Plant Equipment Center ATTN: DIPEC-SSG	b. TELEPHONE (Include Area Code) (1) Commercial: (901)775-6269 (2) AUTOVON: 683-6269
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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